Fabian Espinoza de Osambela

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Education

Louisiana State University 2025–present
Ph.D. in Mathematics
Working with Jerome Hoffman, Fang-Ting Tu, and Scott Baldridge.
Passed qualifying exams in algebra, real analysis, and topology, 2026.
Master's thesis in progress: extended field theories and syntomic invariants of K-trivial covers.

Johns Hopkins University 2025
B.A. in Mathematics and Physics (double major)
General honors; departmental honors in physics.

Research Interests

Algebraic and arithmetic geometry of curves: moduli of curves and of their covers; spin curves, theta characteristics and Prym varieties; class groups and Selmer groups. p-adic Hodge theory: syntomic and prismatic cohomology. Homotopy theory: functorial field theories; motivic homotopy theory and hermitian K-theory; trace methods (topological Hochschild and cyclic homology); formal moduli problems.

Papers

1. F. Espinoza, M. S. Im, and M. Khovanov, *Two-dimensional topological quantum field theories of rank two over Dedekind domains*, Contemp. Math. **834** (2026), 33–52. arXiv:2502.04502.

In preparation

2. F. Espinoza de Osambela, *K-trivial covers of curves and twisted Frobenius algebras.* Expected November 2026.
3. F. Espinoza de Osambela, *K-trivial quadruple covers of curves and their moduli.* Expected December 2026.
4. F. Espinoza de Osambela and S. Jia, *Twisted Frobenius algebras over number rings: Steinitz classes, cyclotomic towers and Selmer groups.* Expected March 2027.
5. F. Espinoza de Osambela, J. Hoffman, and F.-T. Tu, *Genus-2 curves with elliptic subcovers.*

Expository Notes

- *Automorphic forms on the Heisenberg group.* Course project, revised 2026.
- *Spheres and homotopy theory.* Course project, revised 2026.
- *The Grothendieck spectral sequence.* Course project, revised 2026.

Available at fabes124.github.io/publications.

Research Experience

K-trivial covers of curves and genus-2 curves 2025–present
Louisiana State University. With Jerome Hoffman, Fang-Ting Tu, and Scott Baldridge

Finite flat covers of curves with trivial dualizing sheaf, equivalently Frobenius algebra sheaves with trace valued in the canonical bundle: structure theorem, classification and moduli in degree four, and the arithmetic counterpart over rings of integers. Explicit genus-2 curves with split Jacobians.

Topological quantum field theories over algebraic field extensions 2024–2025
Johns Hopkins University. Mentors: Mikhail Khovanov and Mee Seong Im
Frobenius algebra structures over Dedekind domains via projective modules; led to the published paper above.

Earlier undergraduate research, Johns Hopkins University

- Drinfeld centers of fusion 2-categories and their Morita classes; mentor Thomas Waddleton (Ibou Bah's group) 2024
- Von Neumann algebras in mathematical physics, with de Sitter space as a case study; mentor Ibou Bah 2023
- X-ray emission from Galactic black holes and its detectability by eROSITA; mentor Marc Kamionkowski 2023

Graduate Coursework

Louisiana State University. Algebra I; Real Analysis I; Topology I; Algebraic Topology; Operator Theory, Category Theory; Rational Homotopy Theory; Algebraic Geometry; Harmonic Analysis in Phase Space; Quantum Field Theory.

Johns Hopkins University. Algebra I–II; Algebraic Topology I–II; Complex Variables; Riemann Surfaces; Number Theory I; Algebraic Geometry I–II; Lie Groups and Lie Algebras; Riemannian Geometry II. Physics: Quantum Field Theory I–II; General Relativity; Advanced Particle Theory.

Reading Courses

Louisiana State University

- Deformation theory and formal moduli problems (Lurie, Pridham); class field theory; supersymmetry and Morse theory Fall 2026
- Scheme theory and sheaf cohomology; algebraic number theory (number fields); differential geometry and mathematical physics; fusion 2-categories 2025–2026

Johns Hopkins University

- Algebraic geometry: varieties, schemes, and algebraic surfaces, with Vyacheslav Shokurov Spring 2024-Spring 2025
- Characteristic classes (Milnor), with Akira Tominaga. Directed Reading Program Spring 2023
- Differential geometry (Nakahara), with Milton Lin. Directed Reading Program Fall 2022

Talks and Posters

- *On the combinatorics of symmetric elliptic subcovers of genus 2 curves.* Informal Combinatorics Seminar, Louisiana State University, November 2026.
- *Twisted Frobenius algebras on curves.* Math-Physics Joint Seminar, University of Pennsylvania, November 2026.
- *The Grothendieck spectral sequence.* Informal Geometry/Topology Seminar, Louisiana State University, October 2026.
- *Renormalization group and Ricci flow* (poster). Undergraduate Poster Session, Department of Physics and Astronomy, Johns Hopkins University, April 2025.

- *Topological quantum field theories over rings of integers of number fields.* Knots in Washington, George Washington University, December 2024.
- *Topological quantum field theories over algebraic field extensions.* Fall 2024 Regional Undergraduate Math Research Conference, Towson University, November 2024.
- *Drinfeld center of 2-fusion categories.* Spring 2024 Regional Undergraduate Math Research Conference, Towson University, April 2024.
- *Fractional quantum Hall effect* (poster). Undergraduate Poster Session, Department of Physics and Astronomy, Johns Hopkins University, April 2024.
- *Introduction to gauge theory using classical field theory.* Society of Physics Students, Johns Hopkins University, November 2023.

Honors and Awards

- Huel D. Perkins Doctoral Fellowship, Louisiana State University 2025–present
- Undergraduate teaching award, JHU Mathematics 2025
- General honors and departmental honors in physics, Johns Hopkins University 2025

Teaching

Teaching Assistant, Mathematics, Johns Hopkins University 2023–2025
Calculus III (Fall 2023); Methods of Complex Analysis (Spring 2024, Fall 2024); Differential Equations and Applications (Spring 2025). Led weekly problem-solving sections, wrote quizzes, ran review sessions, and graded. Received the department's undergraduate teaching award.

Learning Assistant, Physics and Astronomy, Johns Hopkins University 2021–2025
Active-learning sections of Physics I (Fall 2021, Fall 2022) and Physics II (Spring 2022, Fall 2023) for physical sciences majors; Electromagnetic Theory I (Spring 2025).

Tutor, GoPeer 2021–2022
AP and college calculus and physics.

Service, Mentoring, and Leadership

- Founder and Director, Physics Undergraduate Reading Program, JHU Society of Physics Students 2023–2025
  Modeled on the math Directed Reading Program, for first- and second-year students preparing for research. Led reading courses in tensor calculus, differential topology, category theory, symmetry in particle physics, string theory, and continuum mechanics.
- Mentor and mentee, JHU physics mentorship program; organized review sessions for Advanced Classical Mechanics and Quantum Mechanics 2023-2025
- Academic mentor to about thirty undergraduates in mathematics and physics (course planning, research interests, finding advisors) 2023–2025
- Vice-President, JHU Math Club 2023–2025
- Treasurer, JHU Society of Physics Students (named a Distinguished Chapter) 2022–2023

Skills

Python, Mathematica, MATLAB, Sage, Magma, PARI.